

HK IMO Training 2020-2021
Problem Set 17
For 6 March 2021



Problem 65. We have two piles with 2000 and 2017 coins respectively. Ann and Bob take alternate turns making the following moves: The player whose turn is to move picks a pile with at least two coins, removes from that pile t coins for some $2 \leq t \leq 4$, and adds to the other pile 1 coin. The players can choose a different t at each turn, and the player who cannot make a move loses. If Ann plays first determine which player has a winning strategy.

Problem 66. Two players A and B play alternatively in a convex polygon with $n \geq 5$ sides. In each turn, the corresponding player has to draw a diagonal that does not cut inside the polygon previously drawn diagonals. A player loses if after his turn, one quadrilateral is formed such that its two diagonals are not drawn. A starts the game.

For each positive integer n , find a winning strategy for one of the players.

Problem 67. Mojtaba and Hooman are playing a game. Initially, Mojtaba draws 2018 vectors with zero sum. Then in each turn, starting with Mojtaba, the player takes a vector and puts it on the plane. After the first move, the players must put their vector next to the previous vector (the beginning of the vector must lie on the end of the previous vector). At last, there will be a closed polygon. If this polygon is not self-intersecting, Mojtaba wins. Otherwise Hooman wins. Who has the winning strategy?

Problem 68. Anna and Berta play a game in which they take turns in removing marbles from a table. Anna takes the first turn. When at the beginning of the turn there are $n \geq 1$ marbles on the table, then the player whose turn it is removes k marbles, where $k \geq 1$ either is an even number with $k \leq \frac{n}{2}$ or an odd number with $\frac{n}{2} \leq k \leq n$. A player wins the game if she removes the last marble from the table.

Determine the smallest number $N \geq 100000$ such that Berta can enforce a victory if there are exactly N marbles on the table in the beginning.